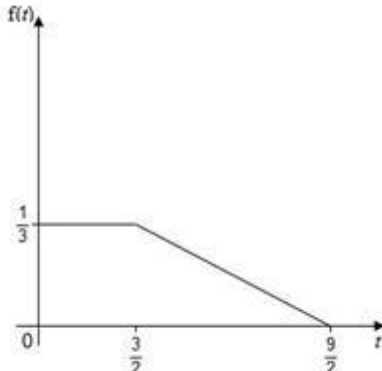


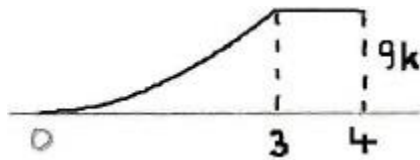
Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Draws horizontal line $(0, \frac{1}{3})$ to $(\frac{3}{2}, \frac{1}{3})$ and straight line joining $(\frac{3}{2}, \frac{1}{3})$ to t axis	AO1.1a	M1	
	Sketches correct shape, accurate and fully labelled	AO1.1b	A1	
(ii)	Deduces median is $\frac{3}{2}$	AO2.2a	B1	Area under $f(t)$ for $0 \leq t \leq \frac{3}{2}$ is $\frac{1}{2}$ Median of t is $\frac{3}{2}$
(b) (i)	States integral required for $E(T)$	AO1.1a	M1	$E(T) = \int_0^{\frac{3}{2}} \frac{1}{3} t dt + \int_{\frac{3}{2}}^{\frac{9}{2}} \frac{t(9-2t)}{18} dt$
	Obtains $E(T)$ correctly	AO1.1b	A1	$= \left[\frac{1}{6} t^2 \right]_0^{\frac{3}{2}} + \left[\frac{t^2}{4} - \frac{t^3}{27} \right]_{\frac{3}{2}}^{\frac{9}{2}} = \frac{13}{8}$
(ii)	Uses definition for variance (PI)	AO1.1a	M1	$\text{Var}(T) = E(T^2) - (E(T))^2$
	Obtains $\text{Var}(T)$	AO1.1b	A1	$\text{Var}(T) = \frac{15}{4} - \left(\frac{13}{8} \right)^2 = \frac{71}{64}$
	Obtains $\text{Var}(4T - 5)$ (Follow through 'their' $\text{Var}(T)$)	AO1.1b	A1F	$\text{Var}(4T - 5) = 16 \text{Var}(T) = \frac{71}{4}$
				Total 8 marks

Q2.

(a)



Curve + rectangle

B1

Some indication of x values

B1

$9k$ or 0.5 indicated for vertical height

B1

3

(b) Attempt to integrate kx^2 between 0 & 3

Show $\frac{kx^3}{3}$

M1

Obtain $9k$

A1

Area under rectangle = $9k$

B1

$9k + 9k = 1$ so $k = \frac{1}{18}$

For $9k + 9k = 1$. AG

B1

4

(c) (i) 3

B1

(ii) Attempt to integrate kx^2 between 0 & Q_1 put = 0.25

M1

$$\frac{(Q_1)^3}{54} = 0.25$$

A1

$$Q_1 = 2.38$$

AWRT or any equivalent exact

$$\text{form } \sqrt[3]{13.5} = \sqrt[3]{\frac{3}{2}(\sqrt[3]{2})} = \sqrt[3]{\frac{3}{2}}(\sqrt[3]{4})$$

A1

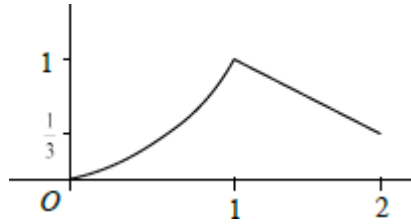
4

[11]



Q3.

(a)



Curve concave upwards between $(0, 0)$ and $(1, y_1)$

B1

Negative gradient line between $(1, y_1)$ and $(2, y_2)$ with $y_2 > 0$ (and not beyond 2)

B1

$y_1 = 1$ and $y_2 = \frac{1}{3}$ shown

B1

3

- (b) (i) Attempt to integrate t^2 **between 0 and x**
Accept integral of x^2

M1

$$F(x) = \frac{1}{3} x^3$$

A1

2

- (ii) Their $F(x) = 0.25$

M1

$$x = 0.909$$

AWRT; accept $\sqrt[3]{0.75}$ OE

A1

2

- (c) (i) $F(1) = \frac{1}{3}$

B1

$$\int_1^x \frac{1}{3}(5-2t) dt = \left[\frac{1}{3}(5t-t^2) \right]_1^x$$

For integral attempted with correct limits

M1

$$= \frac{1}{3}(5x - x^2) - \frac{4}{3}$$

For limits substituted in correct expression

A1

$$F(x) = \frac{1}{3}(5x - x^2) - \frac{4}{3} + \frac{1}{3}$$

$F(1)$ added to give complete $F(x)$

A1

4

$$= \frac{1}{3}(5x - x^2 - 3)$$

AG

$$(ii) \quad \frac{1}{3}(5q - q^2 - 3) = 0.75$$

Setting up equation

or

integral of $f(x)$ from q to $2 = 0.25$

M1

$$4q^2 - 20q + 21 = 0$$

$$\text{or } q^2 - 5q + 5.25 = 0$$

Reaching correct simplified quadratic

A1

$$(2q - 3)(2q - 7) = 0$$

$$\text{or } q = 2.5 \pm 1$$

Factorising for two solutions or using formula or calculator

m1

$$q = 1.5$$

Selecting only this one

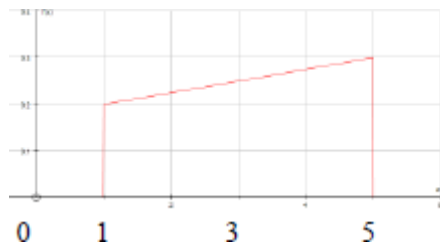
A1

4

[15]

Q4.

(a)



B2 for st. line from (1,0.2) to (5,0.3)

B1 st. line ($m > 0$) from $x = 1$ to $x = 5$.

B2,1

2

(b) $E(X) = \frac{1}{40} \int_1^5 (x+7) dx$
Ignore limits

M1

$$= \frac{1}{40} \left(\frac{x^3}{3} + \frac{7x^2}{2} \right)_1^5$$

Ignore limits

A1

$$= \frac{1}{40} \left(\frac{125}{3} + \frac{175}{2} - \frac{1}{3} - \frac{7}{2} \right)$$

$$= 3\frac{2}{15}$$

cao (accept 3.133 or $\frac{47}{15}$ or **exact**)

A1

3

(c) $F(x) = \int_1^x \frac{1}{40} (x+7) dx$
 $F(x) = \int \left(\frac{x}{40} + \frac{7}{40} \right) dx$

M1

$$= \frac{1}{40} \left[\frac{x^2}{2} + 7x \right]_1^x$$

$$= \frac{x^2}{80} + \frac{7x}{40} + c \Rightarrow (M1A1)$$

A1

$$= \frac{1}{80} (x^2 + 14x - 1 - 14)$$

$$F(1) = 0 \Rightarrow c = -\frac{1}{80} - \frac{7}{40} = -\frac{15}{80}$$

or [use of $F(5) = 1$]

$$= \frac{1}{80} (x^2 + 14x - 15)$$

$$\Rightarrow F(x) = \frac{1}{80} (x^2 + 14x - 15)$$

Adep1

$$= \frac{1}{80} (x + 15)(x - 1)$$

$$F(x) = \frac{1}{80} (x + 15)(x - 1) \text{ (AG)}$$

Adep1

4

(d) (i) $P(2.5 \leq x \leq 4.5) = F(4.5) - F(2.5)$

$$= \frac{1}{80} (19.5 \times 3.5 - 17.5 \times 1.5)$$

Trapezium Rule

$$\frac{1}{2} \left(\frac{23}{80} + \frac{19}{80} \right) \times 2$$

M1

$$= \frac{42}{80} = \frac{21}{40} \text{ (0.525)}$$

A1

2

(ii) $F(m) = \frac{1}{2}$
 $\int_1^m \frac{1}{40} (x + 7) dx = 0.5 \text{ (B1)}$

B1

$$\Rightarrow \frac{1}{80} (m^2 + 14m - 15) = \frac{1}{2}$$

Correct equation formed

M1

$$(\times 80) \Rightarrow m^2 + 14m - 15 = 40$$

$$m^2 + 14m - 55 = 0$$

AG

$$(e) \quad m = \frac{-14 \pm \sqrt{196 + 220}}{2} = \frac{-14 \pm 20.396}{2}$$

Correct attempt at solving quadratic (by formula, oe).

M1

$$m = \frac{-14 + 20.396}{2} \quad (\text{since } m > 1)$$

$$m = 3.198 \text{ (3dp)}$$

cao

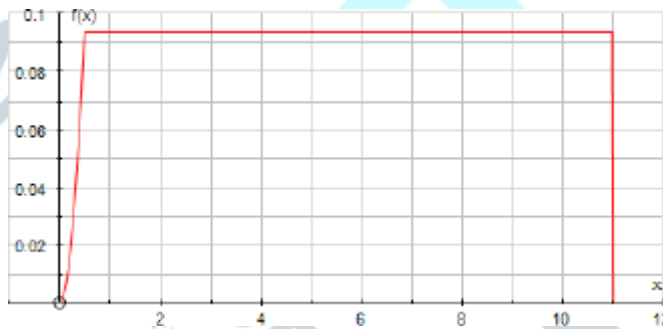
A1

2

[16]

Q5.

(a)



B1 for concave curve from

$(0, 1)$ to $\left(\frac{1}{2}, \frac{3}{32}\right)$

B1 for horizontal straight line

$f = \frac{3}{32}$ from $\left(\frac{1}{2}, \frac{3}{32}\right)$ to $\left(\frac{11}{2}, \frac{3}{32}\right)$

B1 for correct axes

B3

3

$$(b) \quad (i) \quad P\left(X \geq 8\frac{1}{3}\right) = \left[\frac{3}{32} \times \left(11 - 8\frac{1}{3}\right)\right]$$

$$= \frac{3}{32} \times \frac{8}{3}$$

Any correct method attempted in either part

M1

$$= \frac{1}{4}$$

AG

A1

$$(ii) \quad P(X \geq 3) = \frac{3}{32} \times (11 - 3)$$

$$= \frac{3}{4}$$

Any correct method attempted

AG

A1

3

$$(c) \quad (i) \quad \text{Interquartile Range} = 5\frac{1}{3}$$

cao

B1

$$(ii) \quad \text{Median} = 5\frac{2}{3}$$

Cao

B2

Alternative:

$$\frac{1}{64} + \frac{3}{32} \left(m - \frac{1}{2} \right) = \frac{1}{2}$$

$$\Rightarrow 3 \left(m - \frac{1}{2} \right) = 15.5 \Rightarrow m = 5\frac{2}{3}$$

sc if B0 then:

M1 for correct method seen

$$\frac{1}{2} \left(8\frac{1}{3} + 3 \right) \text{ or } \frac{1}{2} \times 11\frac{1}{3}$$

$$\text{or } \frac{3}{32} (11 - m) = \frac{1}{2} \Rightarrow 11 - 5\frac{1}{3}$$

3

$$(d) \quad P[(X < m) \cap (X \geq 3)] = \frac{1}{4}$$

$$\left(\frac{3}{4} - \frac{1}{2} \right)$$

attempted

B1

$$P(X < m \mid (X \geq 3)) = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

(their p) $\frac{1}{3}$ for $0 < p < 1$

M1

cao

A1

Alternative:

(Ratio of relevant two areas)

$$P(X < m \mid X \geq 3) = \frac{2\frac{2}{3}}{8} = \frac{1}{3}$$

cao

3

[12]

Q6.

(a) $F(x) = \int \frac{3}{8}(x^2 + 1) dx$

Ignore limits

M1

$$= \frac{3}{8} \left[\frac{x^3}{3} + x \right] \text{ or } = \frac{1}{8}x^3 + \frac{3}{8}x$$

Either

A1

$$= \frac{1}{8}x(x^3 + 3)$$

(including use of correct limits 0 and x or +c used and evaluated) (AG)

A1

3

(b) $F(m) = \frac{1}{2}$

B1

$$F(1) = \frac{1}{8} \times 1 \times 4 = \frac{1}{2}$$

AG

B1

- (c) Upper quartile lies in range $1 < x < 2$ such that $F(q) = \frac{3}{4}$

$$\frac{1}{2} + \int_1^q \frac{1}{4}(5 - 2x)dx = \frac{3}{4}$$

$$\int_1^q \frac{1}{4}(5 - 2x)dx = \frac{1}{4}$$

Alternative: $\int_q^2 \frac{1}{4}(5 - 2x)dx = \frac{1}{4}$

M1

$$[5x - x^2]_1^q = 1$$

$$[5x - x^2]_q^2 = 1$$

$$5q - q^2 - 4 = 1$$

$$(10 - 4) - (5q - q^2) = 1$$

$$6 - 5q + q^2 = 1$$

$$q^2 - 5q + 5 = 0$$

$$q^2 - 5q + 5 = 0$$

A1

$$q = \frac{5 \pm \sqrt{25 - 20}}{2} \text{ or } \frac{1}{2}(5 \pm \sqrt{5})$$

Correct use of formula (OE) to give the two surd answers to given quadratic equation

M1

but $1 < q < 2$ [or $(q < 2)$]

m1

$$\therefore q = \frac{1}{2}(5 - \sqrt{5})$$

Must qualify with a numerical comparison, not just quote the given answer; dep on M1; AG

A1

(d) $P(X > 1.5) = \frac{1}{2} \left[\frac{1}{2} + \frac{1}{4} \right] \times \frac{1}{2}$

$$P(X < 1.5) = 0.5 + \frac{1}{2} \left[\frac{3}{4} + \frac{1}{2} \right] \times \frac{1}{2} \quad (\text{M1})$$

M1

$$\begin{aligned}
 &= \frac{3}{16}(0.1875) \\
 &= \frac{1}{2} + \frac{1}{2} \times \frac{5}{4} \times \frac{1}{2} \\
 &= \frac{1}{2} + \frac{5}{16} = \frac{13}{16} \quad (\text{A1})
 \end{aligned}$$

A1

$$\begin{aligned}
 P(X > q) &= \frac{1}{4}(0.25) \\
 P(X < q) &= \frac{3}{4}(0.75) \quad (\text{B1})
 \end{aligned}$$

B1

$$\begin{aligned}
 P(q < X < 1.5) &= \frac{1}{4} - \frac{3}{16} \\
 &= \frac{1}{16} \quad (0.0625) \\
 P(q < X < 1.5) &= \frac{13}{16} - \frac{3}{4} = \frac{1}{16} \quad (\text{A1}) \\
 &\quad (0.0625)
 \end{aligned}$$

A1

OR

$$\int_{1.5}^2 \frac{1}{4}(5-2x)dx = \frac{3}{16} \text{ etc } (\text{M1A1})$$

OR

$$\int_q^{1.5} \frac{1}{4}(5-2x)dx = \frac{1}{4} [5x - x^2]_q^{1.5} \quad (\text{M1})$$

(correct integration and limits)

Allow use of $q = 1.38$ to $q = 1.382$ in limits for M1

Whatever follows **must** be **exact**

$$= \frac{1}{4} [(7.5 - 2.25) - (5q - q^2)] \quad (\text{A1})$$

for use of $5q - q^2 = 5$ or showing

$$5q - q^2 = 5 \text{ by substituting } q = \frac{1}{2}(5 - \sqrt{5}) \quad (\text{A1})$$

$$= \frac{1}{4} [5.25 - 5] = \frac{1}{16} \quad (\text{A1})$$

NB statement

$$F(1.5) - \frac{3}{4} = \frac{1}{16} \quad (\text{OE})$$

scores 4 marks

Alternative:

$$\begin{aligned}\int_q^{1.5} \frac{1}{4}(5-2x)dx &= \left[-\frac{1}{16}(5-2x)^2 \right]_{\frac{5-\sqrt{5}}{2}}^{1.5} \quad (\text{M1}) \\ &= -\frac{1}{16}(4) - \left[-\frac{1}{16}(\sqrt{5})^2 \right] \quad (\text{sub}) \quad (\text{A1}) \\ &= -\frac{4}{16} + \frac{5}{16} \quad (\text{A1}) \\ &= \frac{1}{16} \quad (\text{A1})\end{aligned}$$

Alternative using $F(x)$:

for $1 \leq x \leq 2$

$$\begin{aligned}F(x) &= \frac{1}{2} + \int_1^x \frac{1}{4}(5-2x)dx \\ &= \frac{1}{2} + \frac{1}{4}[5x - x^2]_1^x \\ &= \frac{1}{2} + \frac{1}{4}[(5x - x^2) - (5 - 1)] \\ &= \frac{1}{4}(2 + 5x - x^2 - 4) \\ &= \frac{1}{4}(5x - x^2 - 2) \quad (\text{seen or used}) \quad (\text{M1}) \\ F(1.5) &= \frac{1}{4}(7.5 - 2.25 - 2) = \frac{3.25}{4} \\ &= 0.8125 = \frac{13}{16} \quad (\text{A1}) \\ F(q) &= \frac{1}{16}(50 - 10\sqrt{5} - (25 - 10\sqrt{5} + 5) - 8) \\ &= \frac{12}{16} \quad \text{OE} \quad (\text{B1}) \\ P(q < X < 1.5) &= \frac{13}{16} - \frac{12}{16} = \frac{1}{16} \quad (\text{A1})\end{aligned}$$

[14]

Q7.

(a) Median = 1

B1

$$\text{Lower quartile} = \frac{1}{2}$$

B1

2

$$\frac{1}{2}$$

(b) $F(1) =$

For $1 \leq x \leq 4$

$$\int \frac{1}{18} (x-4)^2 dx$$

ignore limits

M1

$$= \left[\frac{1}{54} (x-4)^3 \right]_1^x$$

$$= \left[\frac{1}{54} (x-4)^3 + \frac{1}{2} \right]$$

Correct integration + correct limits seen or used

A1

$$F(x) = \left[\frac{1}{54} (x-4)^3 + \frac{1}{2} \right] + \frac{1}{2}$$

adding $\frac{1}{2}$ or $F(1)$

m1

$$= 1 + \frac{1}{54} (x-4)^3$$

CAO (AG)

A1

Alternative

$$\int \frac{1}{18} (x-4)^2 dx = \frac{1}{54} (x-4)^3 + c$$

(M1)

$$F(1) = \frac{1}{2} \Rightarrow c = 1$$

(m1)(A1)

$$F(x) = 1 + \frac{1}{54} (x-4)^3$$

(A1)

Alternative

$$\int \frac{1}{18} (x-4)^2 dx \quad (M1)$$

$$= \int_1^x \frac{1}{18} (x^2 - 8x + 16) dx$$

$$= \frac{1}{18} \left[\frac{x^3}{3} - 4x^2 + 16x \right]_1^x \quad (A1)$$

$$F(x) = \frac{1}{2} + \frac{1}{54} [x^3 - 12x^2 + 48x]_1^x \quad (M1)$$

$$= \frac{1}{2} + \frac{1}{54} (x^3 - 12x^2 + 48x - 37)$$

$$= 1 + \frac{1}{54} (x^3 - 12x^2 + 48x - 64)$$

$$= 1 + \frac{1}{54} (x-4)^3 \quad (A1)$$

$$\frac{53}{54} - \frac{46}{54}$$

$$F(3) - F(2)$$

$$(c) \quad P(2 \leq X \leq 3) =$$

M1

$$= \frac{7}{54} (0.130)$$

$$0.1296$$

A1

2

(d) (i)

$$F(q) = \frac{3}{4}$$

$$\text{use of } F(q) = \frac{3}{4}$$

M1

$$1 + \frac{1}{54} (q-4)^3 = \frac{3}{4}$$

$$\frac{1}{54} (q-4)^3 = -\frac{1}{4}$$

(either)

M1

$$(\times 54) \Rightarrow (q-4)^3 = -13.5$$

AG

A1

3

(ii) $q - 4 = \sqrt[3]{-13.5} = -2.3811 \quad q = 1.619 \text{ (3dp)}$

CAO

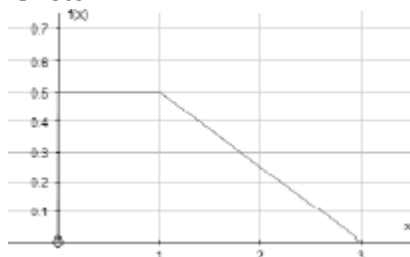
B1

1

[12]

Q8.

(a) Sketch:



1 for straight line $0 \leq x \leq 1$ from $(0, 0.5)$ to $(1, 0.5)$

1 for straight line $1 \leq x \leq 3$ from $(1, 0.5)$ to $(3, 0)$

1 for axes [must have at least $(0, 0.5)$
 $(1, 0)$ and $(3, 0)$ labelled]

B3

3

(b) $P(X \leq \eta) = F(\eta) = 0.5$

M1

$(\Rightarrow \eta = 1 \text{ (from graph)})$

AG

A1

2

(c) $\mu = E(X) = \int_0^1 \left(\frac{x}{2}\right) dx + \int_1^3 x \left(\frac{3-x}{4}\right) dx$

Both integrals stated

M1

$$= \left[\frac{x^2}{4} \right]_0^1 + \frac{1}{4} \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_1^3$$

Either

A1

$$= \frac{1}{4} + \frac{1}{4} \left[\left(\frac{27}{2} - 9 \right) - \left(\frac{3}{2} - \frac{1}{3} \right) \right]$$

Correct limits used on both

m1

$$= \frac{1}{4} + \frac{5}{6} \quad (0.25 + 0.83\dot{3})$$

$$= 1\frac{1}{12}$$

(CAO)

A1

4

(d) Area of Δ

$$= P\left(X > 2\frac{1}{4}\right) = \frac{1}{2} \times \frac{3}{4} \times \frac{3 - 2\frac{1}{4}}{4}$$

M1ft

$$= \frac{3}{32} \times \frac{3}{4} = \frac{9}{128}$$

$$\therefore P\left(X < 2\frac{1}{4}\right) = 1 - \frac{9}{128}$$

M1ft

$$= \frac{119}{128} (0.9296875)$$

A1

Alternative:

For $1 \leq x \leq 3$

$$F(x) = 1 - \frac{1}{8}(3-x)^2 \quad \text{M1ft}$$

$$F\left(2\frac{1}{4}\right) = 1 - \frac{1}{8} \times \frac{9}{16} \quad \text{M1ft}$$

$$= \frac{119}{128}$$

CAO

or

$$\int_{2\frac{1}{4}}^3 \frac{3-x}{4} dx \left(= \frac{9}{128} \right)$$

M1ft

$$= 1 - \int_{2\frac{1}{4}}^3 \frac{3-x}{4} dx$$

M1ft

$$\begin{aligned} &= 1 - \frac{1}{4} \left[3x - \frac{x^2}{2} \right]_{2\frac{1}{4}}^3 \\ &= 1 - \frac{1}{4} \left[9 - \frac{9}{2} - \frac{27}{4} + \frac{81}{32} \right] \\ &= 1 - \frac{1}{4} \times \frac{9}{32} = \frac{119}{128} \end{aligned}$$

A1

or $(1 - 0.0703125 = 0.9296875)$

Alternative

$$f\left(2\frac{1}{4}\right) = \frac{3}{16} = 0.1875$$

$$P(X < 3\mu - \eta) = P\left(X < 2\frac{1}{4}\right)$$

$$= \frac{1}{2} + \frac{1}{2} \left(\frac{3}{16} + \frac{1}{2} \right) \times 1\frac{1}{4} \quad \text{M1ft}$$

$$= \frac{1}{2} + \frac{55}{128} (0.4296875) \quad \text{M1ft}$$

$$= \frac{119}{128} (0.930) \quad \text{A1}$$

3

[12]

Q9.

(a) $P(X < 0) = F(0)$

M1

$$= \frac{1}{k+1}$$

A1

2

(b) $(q_1 + 1) \times \frac{1}{k+1} = \frac{1}{4}$

alternative (from a sketch)

M1

$$q_1 + 1 = \frac{1}{4}(k+1)$$

A1

$$q_1 = \frac{1}{4}(k+1) - 1$$

OE

A1

3

(c) $f(x) = \frac{d}{dx}(F(x))$

use of

M1

$$\left. \begin{aligned} &= \frac{1}{k+1} \times \frac{d}{dx}(x+1) \\ &= \frac{1}{k+1} \quad -1 \leq x \leq k \\ &= 0 \quad \text{otherwise} \end{aligned} \right\}$$

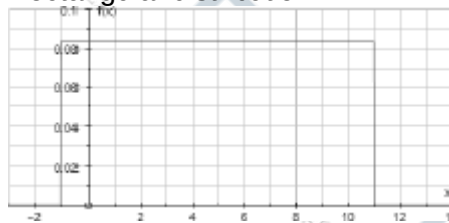
AG; $\frac{1}{k+1}$ clearly deduced

A1

2

(d) (i) $k = 11 \Rightarrow f(x) = \begin{cases} \frac{1}{12} & -1 \leq x \leq 11 \\ 0 & \text{otherwise} \end{cases}$

Rectangular distribution:



horizontal line on $[-1, 11]$

B1

$$\text{at } f = \frac{1}{12}$$

B1

2

(ii) $E(X) = \frac{1}{2}(-1 + 11) = 5$

B1

$$\text{Var}(X) = \frac{1}{12} (11 - (-1))^2 = 12$$

B1

2

$$(iii) \quad P(a_1 < X < E(X)) = P(2 < X < 5)$$

$$= (5 - 2) \times \frac{1}{12}$$

M1

$$= 0.25$$

AG

A1

2

[13]

Q10.

$$\int_0^1 t f(t) dt$$

(a) $E(T) =$

$$= \int_0^1 4t^2 (1 - t^2) dt$$

M1

$$= \left(\frac{4t^3}{3} - \frac{4t^5}{5} \right) \Big|_0^1$$

A1

$$= \frac{4}{3} - \frac{4}{5}$$

A1

$$= \frac{8}{15}$$

AG

3

$$(b) \quad (i) \quad F(t) = P(T < t) = \int_0^t f(t) dt$$

$$\int_0^t 4t (1 - t^2) dt$$

M1

$$= (2t^2 - t^4) \Big|_0^t$$

$$= 2t^2 - t^4$$

A1

2

$$(ii) \quad P(\mu < T < m) = F(m) - F(\mu)$$

$\Downarrow$

M1

$$F(m) = 0.5$$

B1

$$F(\mu) = F\left(\frac{8}{15}\right) = 0.4880$$

B1

$$P(\mu < T < m) = 0.5 - 0.488$$

0.5 - their $F(\mu)$

M1ft

$$\therefore = 0.012$$

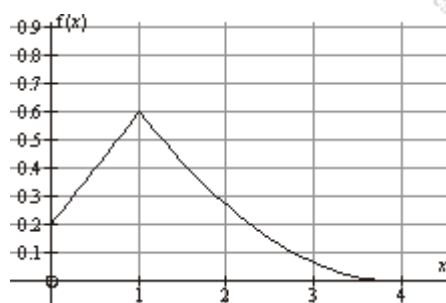
A1

5

[10]

Q11.

(a)



B1 for line segment

(0, 0.2) to (1, 0.6)

B1 for correct shaped curve

(1, 0.6) to (4, 0)

B2

2

(b) (i) for $0 \leq x \leq 1$

$$F(x) = \int_0^x \frac{1}{5} (2x) dx$$

M1

$$= \frac{1}{5} (x^2 + x) \Big|_0^x$$

A1

$$\frac{1}{5} x(x+1)$$

A1

3

(ii) $P(X \geq 0.5) = 1 - F(0.5)$

$$= 1 - \frac{1}{5} \times 0.5 \times 1.5$$

M1

$$= 1 - 0.15$$

$$= 0.85$$

(OE)

A1

2

(iii) $F(q_1) = 0.25$

$$\frac{1}{5} x(x+1) = 0.25$$

↓

B1

$$F(0.5) = \frac{3}{20} = 0.15 < 0.25$$

$$F(0.75) = \frac{21}{80} = 0.2625 > 0.25$$

Attempt to solve

↓

$$q_1 = 0.725$$

⇓

M1

$$\therefore 0.5 < q_1 < 0.75$$

$$0.5 < q_1 < 0.75 \quad \text{AG}$$

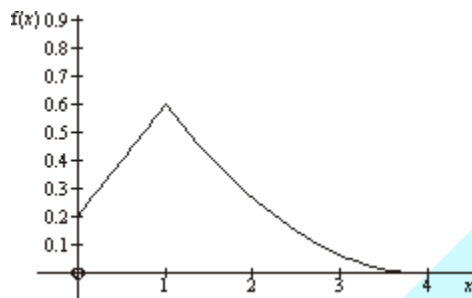
A1

3

[10]

Q12.

(a)



B1 for line segment (0,0.2) to (1,0.6)

B1 for correctly shaped curve (1,0.6) to (4,0)

B2

2

(b) (i) for $0 \leq x \leq 1$

$$F(x) = \int_0^x \frac{1}{5}(2x+1)dx$$

Ignore limits

M1

$$= \left[\frac{1}{5}(x^2 + x) \right]_0^x$$

Ignore limits

A1

$$= \frac{1}{5}x(x+1)$$

A1

$$\begin{aligned} P(X \leq 1) &= F(1) \\ &= \frac{2}{5} \end{aligned}$$

(ii)

B1

1

$$(iii) \quad P(X \geq x) = \frac{17}{20} \Rightarrow F(x) = \frac{3}{20}$$

M1

$$\frac{1}{5} x(x+1) = \frac{3}{20}$$

m1

$$x(x+1) = \frac{3}{4}$$

$$x^2 + x - \frac{3}{4} = 0$$

A1

$$\left(x - \frac{1}{2}\right)\left(x + \frac{3}{2}\right) = 0$$

Any valid method attempted

m1

$$x = \frac{1}{2}$$

A1

5

$$(iv) \quad \text{Since } F(1) = 0.4, q \text{ lies in } 0 \leq r \leq 1$$

$$F(q) = \frac{1}{5} (q^2 + q) = 0.25$$

M1

$$\Rightarrow q^2 + q = 1.25$$

$$q^2 + q - 1.25 = 0$$

A1

$$\Rightarrow q = \frac{-1 \pm \sqrt{1 - 4 \times (-1.25)}}{2}$$

m1

$$q = \frac{1}{2}(\sqrt{6} - 1) \quad (q > 0)$$

AWFW (0.724 to 0.725)

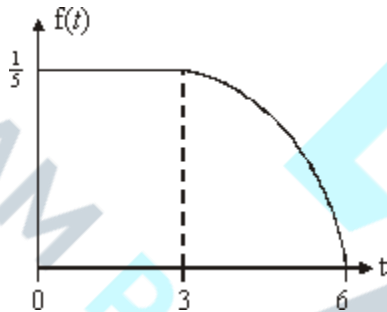
A1

4

[15]

Q13.

(a)



B1 2 axes with scales
B1 Horizontal line at 0.2 from 0 to 3
B1 Curve from 3 to 6

B3

3

(b) $P(T = 3) = 0$

B1

1

(c) $P(T \geq 3) = 1 - P(T < 3)$

M1

$$= 1 - \frac{3}{5}$$

$$= \frac{2}{5}$$

$$\int_3^6 \frac{1}{45} t(6-t) dt = \frac{2}{5}$$

A1

2

$$(d) \int_0^m \frac{1}{5} dt = 0.5$$

$$P(T \leq 3) = 0.6$$

M1

$$\therefore \left(\frac{t}{5}\right)_0^m = 0.5$$

$$\therefore 0 \leq \text{median} < 3$$

$$\frac{m}{5} - 0 = 0.5$$

$$\frac{1}{5}m = 0.5$$

$$m = 0.5 \times 5$$

$$m = 5 \times 0.5$$

$$m = 2.5$$

$$m = 2.5 \quad \text{AG}$$

A1

2

$$(e) \quad E(T) = \int_0^3 \frac{1}{5} t \, dt + \int_3^6 \frac{1}{45} t^2 (6-t) \, dt$$

M1

$$= \left[\frac{1}{10} t^2 \right]_0^3 + \left[\frac{2}{45} t^3 - \frac{1}{180} t^4 \right]_3^6$$

A1A1

$$= \frac{9}{10} + 1.65$$

$$= 2.55$$

A1

$$\therefore P(\text{median} < T < \text{mean})$$

$$= P(2.5 < T < 2.55)$$

M1

$$= 0.05 \times \frac{1}{5}$$

$$= 0.01$$

A1

6

[14]

Q14.

(a) $k = 0.1$

OE.

B1

1

(b) $E(X) = 1$

B1

1

(c) $P(X > 0) = 6 \times 0.1$

M1

$$= 0.6$$

A1

2

(d) $P(|X| > 3.5) = 1 - P(|X| < 3.5)$

M1

$$= 1 - 0.7$$

A1

$$= 0.3$$

A1

3

Alternative solution

$$P(X < -3.5) + P(X > 3.5)$$

(M1)

$$= \frac{0.5}{10} + \frac{2.5}{10}$$

(A1)

$$= \frac{3}{10}$$

(A1)

[7]

Q15.

(a) $E(X) = 4 \quad E(X^2) = 17.2$

$\text{Var}(X) = E(X^2) - (E(X))^2$
use of

M1

$= 17.2 - 4^2$
 $= 1.2$

CAO

A1

2

(b) $C = 2\pi(X + 8)$

OE
Either

or $E(X + 8) = 12$
 CAO

B1

Thus $E(C) = 24\pi$
 CAO

B1

$\text{Var}(C) = 4\pi^2 \times \text{Var}(X)$
use of $V(aX + b) = a^2 \times V(X)$
with $a > 1$ and $b > 0$

M1

Thus $\text{Var}(C) = 4.8\pi^2$
ft on $V(X)$; but must include π^2

A1ft

4

(c) (i) Area, $S = \pi(X + 8)^2$
use of πr^2

M1

$= \pi(X^2 + 16X + 64)$

Thus $a = 16$ and $b = 64$
 CAO both

A1

2

(ii) $E(S) = \pi(E(X^2) + 16E(X) + 64)$

attempted application of E to expanded expression in (c)(i)
 M0 for use of $E(X^2) = 4^2 = 16$

M1

$$= \pi (17.2 + 64 + 64) = 145.2\pi$$

CAO

A1

2

[10]

Q16.

(a) $P(T = 4) = 0$

B1

1

(b) median = 3 $\Rightarrow P(T \leq 3) = 0.5$

$$P(T > 3) = \frac{1}{2} \times f(3) \times (5 - 3)$$

M1

$$= \frac{1}{2} \times 0.5 \times 2$$

$$= 0.5$$

A1

2

(c) $P(T \geq 3.5) = \frac{1}{2} \times f(3.5) \times (5 - 3.5)$

M1

$$= \frac{1}{2} \times \frac{3}{8} \times 1.5$$

M1A1

$$= \frac{9}{32}$$

$$0.28125$$

A1

$$\therefore P(T < 3.5) = 1 - \frac{9}{32}$$

$$= \frac{23}{32}$$

0.71875 (AWRT 0.719)

A1

5

$$(d) \quad E(T) = \int_0^3 \frac{t^3}{18} dt + \frac{1}{4} \int_3^5 t(5-t) dt$$

M1

$$= \left[\frac{t^4}{72} \right]_0^3 + \left[\frac{1}{4} \left\{ \frac{5t^2}{2} - \frac{t^3}{3} \right\} \right]_3^5$$

Integration attempted

M1A1

$$= \frac{9}{8} + \frac{1}{4} \left\{ \left(\frac{125}{2} - \frac{125}{3} \right) - \left(\frac{45}{2} - 9 \right) \right\}$$

Substituting limits

M1

$$= 1 \frac{1}{8} + 1 \frac{5}{6}$$

$$= 2 \frac{23}{24}$$

2.9583 (AWRT 2.96)

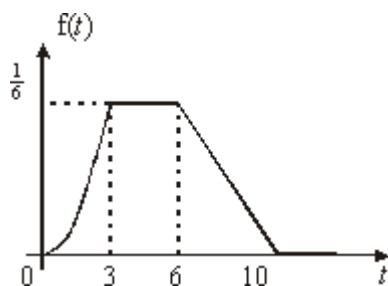
A1

5

[13]

Q17.

(a)



1 for each correct part of graph

B4

4

1 for axes $\left(\frac{1}{6} \text{ and } 10; f(t); t \right)$

(b) (i) $P(3 \leq T < 6) = \text{Area of rectangle}$

M1

$$= 3 \times \frac{1}{6}$$

$$= \frac{1}{2}$$

A1

2

(ii) $P(T > 6) = \text{Area of triangle}$

M1

$$= \frac{1}{2} \times \frac{1}{6} \times 4$$

$$= \frac{1}{3}$$

(accept 0.33)

A1

2

(iii) $\therefore P(T < 3) = 1 - \left(\frac{1}{3} + \frac{1}{2} \right) = \frac{1}{6}$

(seen anywhere) or $\int_0^3 f(t) dt = \frac{1}{6}$

B1

$$P(T < \text{lower quartile}) = \frac{1}{4}$$

$$P(T > Q_1) = \frac{3}{4} \text{ or } \int_0^{Q_1} f(t) dt = \frac{1}{4}$$

M1

$\therefore$ lower quartile lies in the range $3 < T < 6$

$\therefore$ lower quartile $= 3 + d$

$$\text{Area} = d \times \frac{1}{6} = \frac{1}{4} \times \frac{1}{6} = \frac{1}{12}$$

Any valid attempt at d

M1

$$\text{where } d = \frac{1}{12} \div \frac{1}{6} = \frac{1}{2}$$

$$\therefore \text{lower quartile} = 3 + \frac{1}{2} = 3.5$$

CAO

A1

4

(c) $P(T < 2) = \int_0^2 \left(\frac{1}{54} t^2 \right) dt$
(ignore limits)

M1

$$= \left\{ \frac{1}{162} t^3 \right\}_0^2$$

(for $\frac{1}{162} t^3$; OE)

A1

$$= \frac{4}{81}$$

$$= 0.0494 \text{ (3sf)}$$

$$= 5\%$$

$$= \frac{4}{81}$$

$$= 0.0494 \text{ (3sf)}$$

$$\approx 5\%$$

(Substitution of correct limits)

m1

any correct form (AWRT 0.049)

A1

4

[16]

Q18.

(a) (i) $f(t) = 2e^{-2t}, \quad (t \geq 0)$
no penalty for omitting $t \geq 0$

B1

1

(ii) Mean = 0.5
CAO

B1

Standard deviation = 0.5
CAO

B1

2

(b) $P(T > 3) = 1 - F(3)$

$= 1 - (1 - e^{-6})$
or by integration

M1

$= 0.0248$
AWRT 0.00248

A1

2

(c) Median m satisfies $F(m) = 0.5$
 $1 - e^{-2m} = 0.5$
or by integration

M1

$e^{-2m} = 0.5$
 $-2m = \ln(0.5)$
or $e^{2m} = 2$

$m = -0.5 \ln(0.5)$
giving $0.5 \ln(2)$

A1

$(= 0.347)$

Median delay = 20.82
= 21 seconds to nearest second
CAO

A1

3

[8]

Q19.

(a) $\int_1^5 ks^2 ds = 1$
for $\int ks^2 ds$

M1

$\Rightarrow \left[\frac{ks^3}{3} \right]_1^5 = 1$

$$\text{for } \frac{ks^3}{3}$$

A1

$$\Rightarrow \frac{124}{3}k = 1$$

$$k = \frac{3}{124}$$

AG

A1

4

$$(b) \quad P(S < 3) = \frac{3}{124} \int_1^3 s^2 ds$$

$$1 - \int_3^5 ks^2 ds$$

M1

$$= \left[\frac{s^3}{124} \right]_1^3$$

(limits must be correct for M1)

$$= \frac{27}{124} - \frac{1}{124}$$

$$= \frac{26}{124}$$

$$\text{substitute limits } 1 - \left(\frac{125}{3} - \frac{27}{3} \right) k$$

$$= 1 - \frac{49}{62} = \frac{13}{62}$$

m1

$$= \frac{13}{62}$$

AWRT 0.210

A1

3

$$(c) \quad \frac{3}{124} \int_1^m s^2 ds = 0.5$$

$$\int_k^5 s^2 ds = 0.5$$

M1

$$\left[\frac{s^3}{124} \right]_1^m = 0.5$$

$$\left[\frac{s^3}{124} \right]_m^5 = 0.5$$

A1

$$m^3 - 1 = 62$$

$$125 - m^3 = 62$$

M1

$$m^3 = 63$$

$$m = \sqrt[3]{63}$$

$$m = 3.98 \text{ (3SF)}$$

$$m = 4 \text{ min}$$

A1

4

[11]

Q20.

(a) (i) 1

1 cao

B1

(ii) 1.54

1.54 cao

B1

(iii) 1.56

1.56 cao

B1

(iv) 1.73

1.73 cao

B1

(v) 1

1 cao

B1

5

(b) (i) $0.5 - 0.291 = 0.209$
method

M1

0.209 cao

A1

2

(ii) $\sqrt{0.0751} = 0.2474$
method

M1

$0.274 \text{ (0.2735 to 0.2745)}$

A1

2

allow B1 for variance = 0.0751

any correct equation

(iii) $E(X^2) - 1.54^2 = 0.0751$

M1

$E(X^2) = 2.45$
 $2.45 \text{ (2.44 to 2.45)}$

A1

2

[11]

Q21.

(a) $\int_0^2 cx \, dx = 1$

Any correct expression - ignore limits

M1

$\left[\frac{cx^2}{2} + dx \right]_0^2 = 1$

Any correct integration - may be awarded elsewhere in question

M1

$2c + 2d = 1$

Wholly correct proof allow geometrical proof

A1

(b) (i) $\int_0^m 0.5 \, dx = 0.5$
Any correct expression

M1

$[0.5x]_0^m = 0.5$ $m = 1$
1 **cao** by correct method allow geometrical proof

A1

2

(ii) $\int_0^m 0.5x \, dx = 0.5$ $m = 1$
Any correct expression

M1

$\left[\frac{0.5x^2}{2}\right]_0^m = 0.5$
Any correct equation after integration

M1

$m = \sqrt{2} = 1.41$
 $\sqrt{2}$ or 1.41 (1.41 to 1.42) disallow $\pm \sqrt{2}$

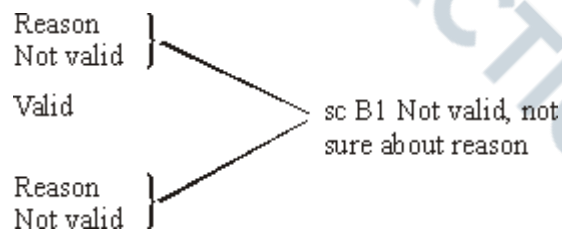
A1

3

- (c) **A** not valid - $2c + 2d \neq 1$
B valid
C not valid = $f(x)$ negative for small x

M1

5



A1

B1

M1

Q22.

- (a) Maximum of p.d.f.

Maximum of graph

E1

1

- (b) (i) Variance = $0.369 - 0.542^2 = 0.075236$

Method for variance

M1

$$\text{s.d.} = \sqrt{0.075236} = 0.274$$

Method for s.d. - disallow if called variance

m1

0.274 (0.274 – 0.275)
SC allow A1 for variance =
 0.0752 (0.0752 – 0.0753)

A1

3

- (ii) $1 - 0.719 = 0.281$

Use of 0.719 0.281 cao

B1B1

2

- (iii) 0.557

0.557 cao

B1

1

- (iv) $0.5 - 0.483 = 0.017$

Use of 0.483, allow $1 - 0.483$

B1

Correct method

M1

0.017 cao

A1

3

Examiner reports

Q2.

This was another well-answered question, with many students scoring full marks. A plot was not required for the sketch (though many did produce one) but an indication of scale on both axes was required. Students should appreciate the distinction between a requirement to 'Show' as in part (b), where each step of the working is expected, and 'Write down' as in part (c)(i) where only the answer is needed. Having shown that the area splits neatly in half as $9k + 9k$, many students wasted time with up to a page of working to find the median value.

Q3.

The sketch was reasonably well done and most students correctly showed the integration from 0 to x to find $F(x)$. The calculation of the lower quartile was also well done. In part (c)(i), there was much fudging of the answer after the integral from 1 to x gave -4 while the required answer was -3 , causing many to lose the mark for the correctly evaluated integral when they spuriously altered it. Often an incorrect answer to part (i) was carried over into part (ii) giving an incorrect equation, with the given $F(x)$ been ignored. It was disconcerting to see how many students at this level could not even recognise or subsequently solve a quadratic equation. On the other hand, many students scored highly on all parts of this question.

Q4.

Candidates need to realise that when asked to 'sketch' a given function, it need not be accurate but that the important features must be shown accurately. This time, fewer but still too many freehand drawings were seen, and some candidates did not give enough detail to gain full credit as they did not take enough care in making sure that the line segment joined $(1, 0.2)$ to $(5, 0.3)$ and was straight. Part (b) was often very well done, with many candidates gaining full marks. However, some did not give their answer in the exact form asked for and consequently lost a mark. It was disappointing to see some treating the distribution of X as discrete. Their attempt to find $E(X)$, by constructing a table of values of p for $x = 1, 2, 3, 4$ and 5 gained no marks. In part (c), those candidates who

correctly stated $F(x) = \int_1^x \frac{1}{40} (x + 7) dx$, often managed to complete the proof successfully. Those who ignored these limits but correctly stated that $F(x) =$

$\int \frac{1}{40} (x + 7) dx = \frac{x^2}{80} + \frac{7x}{40} + C$ were less successful in arriving at the given answer. When this route was used, the calculation of the value of C by a valid method must be seen by using either $F(1) = 0$ or $F(5) = 1$ for full marks to be available. Working backwards from the given answer was not considered to be a valid method for finding the value of C . Candidates should realise that, when the answer is given, a full and comprehensive method must be shown in order to be able to achieve full credit. Part (d) was usually completed successfully, with most candidates able to solve the given quadratic equation in part (e). Again, some failed to comply with the instruction 'give your answer to three decimal places' and so lost a mark.

Q5.

The graph in part (a) was usually adequately sketched. There was, however, a minority of candidates who interpreted the request for a 'sketch' as an excuse to offer up a very poor quality freehand drawing. This should not have been the case and any such attempts were likely to be penalised.

The majority of candidates used simple areas of rectangles to obtain their very efficient solutions to parts (b)(i) and (b)(ii). Most candidates realised that, when the answers are given in the question, sufficient working must be shown in order to obtain full credit. Some candidates, when they are confronted by a continuous distribution, seem to think that integration methods always have to be used. This is not the case when straight lines form part of the graph of the probability density function. Those candidates who used valid integration methods usually gained the correct given values and thus full credit.

Candidates were expected to use the information given in part (b) to write down the answers to parts (c)(i) and (c)(ii).

Although some candidates recognised that the answers to part (b) implied that the upper

quartile of X was $8\frac{1}{3}$ and that the lower quartile of X was 3, many attempted to recalculate these values, usually incorrectly, and mostly by the use of calculus methods.

Disappointingly, in part (d), only the most able candidates recognised that, whatever the values obtained for the median and lower quartile, $P(X < \text{median} \mid X \geq \text{lower quartile}) = \frac{1}{3}$ and $P(X < \text{median and } X \geq \text{lower quartile}) = \frac{1}{4}$ and are always true.

Q6.

Although most candidates managed to correctly integrate $\frac{3}{8}(x^2 + 1)$ in part (a), many then failed to use the correct limits required to find $F(x)$, with the vast majority failing to consider any limits at all but simply stating the given answer.

In part (b), many candidates stated $F(m) = 0.5$ but then failed to verify that $m = 1$ satisfied this condition. Part (c) was not well answered except by the more able candidates. Many

started from the incorrect assumption that $\int_0^q \frac{1}{4}(5 - 2x)dx = \frac{3}{4}$. Consequently, the given equation was often quoted without any foundation at the end of a lot of spurious work.

It was also very disappointing to see candidates on this paper unable to solve the given quadratic equation. Those who did manage to obtain two surd answers were often then unable to give an adequate numerical comparison in order to justify the selection of the given answer. Most candidates simply wrote down the given answer without any justification whatsoever. This lost them the final two marks for this part of the question. Part (d), although relatively straightforward, confounded many candidates as very few realised that all they had to do was to evaluate $F(1.5) - F(q)$ where $F(q) = \frac{3}{4}$ and $F(1.5) = \frac{13}{16}$.

Q7.

Part (a) was answered very poorly by too many candidates. Candidates were asked to **state** values for the median and lower quartile of X . This should have indicated to candidates that the answers required little work, but this was often not picked up. Some candidates obtained the correct values but these were often invalidated by incorrect

methods. Those candidates who employed calculus often had more success, as they realised that $F(m) = 0.5$ and $F(q_1) = 0.25$ were required.

Part (b) was often done well by those candidates who realised that $F(x) = 1/54(x - 4)^3 + c$, as they usually went on to use either $F(1) = 1/2$ or $F(4) = 1$ to show that $c = 1$. Some candidates, having been given the answer in the question, simply wrote this down at the end of what was usually a totally incorrect method.

Part (c) was usually done by correctly attempting to evaluate $F(3) - F(2)$ but the arithmetic then defeated many.

Most candidates realised that $F(q) = 3/4$ was required and consequently went on to gain full marks for part (d)(i). The correct answer to part (d)(ii) was often seen, even by those candidates who could not do part (d)(i). However, some failed to give the answer to three decimal places and so lost the mark.

Q8.

In part(a), too many candidates seemed to have the wrong perception that the request for a 'sketch' entitled them to draw a very poor quality graph or copy some unscaled diagram which they had managed to take from the display on their graphics calculators. All such attempts lost some or all of the marks available. The axes and straight lines should have been drawn with the aid of a ruler and should have contained the appropriate scales: O or $(0, 0)$, $(1, 0)$ and $(3, 0)$ on the x -axis and $(0, 0.5)$ on the y -axis as a bare minimum. A straight line from $(0, 0.5)$ to $(1, 0.5)$ and a straight line from $(1, 0.5)$ to $(3, 0)$ should then have been drawn. It was often quite difficult to decide whether the sketch contained the required 'straight' lines or curves, so badly attempted were the sketches.

Similarly, part (b) was very poorly attempted with most candidates simply showing that $f(1) = 0.5$ or solving $f(x) = 0.5$ to give $x = 1$. Both of these were inappropriate and so gained no credit.

There were many good attempts at finding $E(X)$ in part (c). However when, as here, the answer is given in the question, sufficient relevant working must be shown for full credit to be obtained.

Although there were many excellent solutions to part (d), there were only a few candidates

who used the most efficient method of $1 - \frac{1}{2} \times 0.75 \times \frac{3}{16}$; most chose an alternative method which required integration. There were a large number of usually less able

candidates who wrote $P\left(X > 2\frac{1}{4}\right) = f\left(2\frac{1}{4}\right) = \frac{3}{16}$, thus gaining no marks.

Q9.

In part (a), $F(0) = \frac{1}{k+1}$ was all that was required, but the use of $F(0) - F(-1) = \frac{1}{k+1}$ was not penalised. In part (b), most candidates realised that they needed to use the fact that $F(q_1) = 0.25$ and so went on to determine the correct expression for q_1 . Unfortunately many candidates must have thought that k was the lower quartile since they found an expression for k in terms of x . In part (c), candidates were expected to use the relationship

$f(x) = \frac{d}{dx}(F(x))$ in order to gain the required given answer. Unfortunately, attempts at

differentiating $F(x) = \frac{x+1}{k+1}$ were, in the main, unconvincing with little or no attention paid to $F(x) = 0$ and $F(x) = 1$. In part (d)(i), most candidates gained full marks. Many candidates used integration methods in part (d)(ii) which, although correctly done, did waste a lot of time, especially as the answer for $E(X) = 5$ could have been written down by simply

considering the sketch drawn in part (d)(i). Similarly, $\text{Var}(X) = \frac{1}{12} (11 - (-1))^2$ was a more efficient method of finding the variance. There were many well-explained solutions to part (d)(iii) with most candidates realising that $P(2 < X < 5)$ was required. Many candidates correctly stated that $P(X < q_1) = 0.25$ but then assumed that $P(X < E(X)) = 0.5$ without any explanation. This statement was true here since, for a rectangular distribution, the median and the mean take the same value.

Q10.

Candidates seemed well able to attempt part (a) with the result that many successful outcomes were seen. Part (b)(i) was only done correctly by the most able students, with

$\int_0^1 4t(1-t^2) dt$ rarely seen otherwise. Many candidates either used 0 and 1 as their limits or simply ignored limits altogether. It was not sufficient to simply consider the indefinite integral $\int 4t(1-t^2) dt$ without consideration of the boundary conditions.

In part (b)(ii), many candidates realised that $P(\mu < T < m) = F(m) - F(\mu)$ but then went on to use the equation $2m^2 - m^4 = 0.5$ to show that $m = 0.541$ before attempting to

evaluate $\int_{\frac{8}{15}}^{0.541} f(t) dt$, instead of simply evaluating $0.5 - F\left(\frac{8}{15}\right)$.

Q11.

In part (a), candidates showed poor ability in sketching graphs. Common faults were the omission of essential values on both axes, lines drawn as incorrectly passing through the origin on the interval $[0, 1]$ and the quadratic curve incorrectly drawn as convex on the interval $[1, 4]$. In part (b)(i), the vast majority of candidates either ignored the required limits of 0 and x altogether, or used limits of 0 and 1. Consequently, although most stated the required answer, many lost a mark. Part (b)(ii) was usually completed correctly with the correct answer of 0.85 often seen. Part (b)(iii) was not done well. Very few candidates tried to 'verify' the given result. Instead, most attempted to calculate q_x so that they could then show that its value fell within the given interval.

Q12.

In part (a), candidates showed poor ability in sketching graphs. Common faults were the omission of essential values on both axes, lines drawn as incorrectly passing through the origin on the interval $[0, 1]$ and the quadratic curve incorrectly drawn as convex on the interval $[1, 4]$. In part (b)(i), the vast majority of candidates either ignored the required limits of 0 and x altogether, or used limits of 0 and 1.

Consequently, although most stated the required answer, many lost a mark. Part (b)(ii) was usually completed correctly. Several candidates wasted time in part (b)(iii) by considering the probability density function for $x > 1$ with the inevitable loss of most, if not

all, of the available marks. Those candidates who did obtain a quadratic equation usually managed to solve it correctly and nearly always realised that one of their solutions was inadmissible. Part (b)(iv) was often done well, even by those candidates who failed to appreciate what was required in part (b)(iii).

Q13.

Statistics 2A

In general, candidates seemed to have less success with this question. In part (a), the correct graph was only seen very infrequently, with the majority of candidates drawing straight lines to represent both parts of the graph. Where a curve was drawn from $t = 3$ to $t = 6$, it was often concave rather than convex. The vast majority of candidates correctly gave the answer of 'zero' to part (b). However, a minority treated this as a discrete

distribution and used $P(X = 3) = P(X \leq 3) - P(X \leq 2) = \frac{1}{5}$. Part (c) was usually done well. However, many candidates used the more difficult integration route and, the correct answer of 0.4 was often seen. Part (d) was usually attempted quite well with most

candidates aware that the median value, m , had to satisfy $\int_0^m \frac{1}{5} dt = 0.5$. Part (e) was found difficult by all but the more able candidates. Calculating the mean to be 2.55 using the integration of two functions for the two different ranges was beyond most candidates, with several attempting to integrate one or other of the functions between 0 and 6. The final part, calculating $P(\text{median} < T < \text{mean})$ was beyond almost all but the best candidates.

Statistics 2B

In general, candidates seemed to have less success with this question than others on the paper. In part (a), the correct graph was seen very infrequently, with the majority of candidates drawing straight lines to represent both parts of the graph. Where a curve was drawn from $t = 3$ to $t = 6$, it was often concave rather than convex in nature. The great majority of candidates correctly gave the answer of 'zero' to part (b). However, a minority

treated this as a discrete distribution and used $P(X = 3) = P(X \leq 3) - P(X \leq 2) = \frac{1}{5}$. Part (c) was usually done well. However, many candidates used the more difficult integration route and, the correct answer of 0.4 was often seen. Part (d) was usually attempted quite

well with most candidates aware that the median value, m , had to satisfy $\int_0^m \frac{1}{5} dt = 0.5$. Part (e) was found difficult by all but the more able candidates. Calculating the mean to be 2.55 using the integration of two functions for the two different ranges was beyond most candidates, with several attempting to integrate one or other of the functions between 0 and 6. The final part, calculating $P(\text{median} < T < \text{mean})$ was beyond almost all but the best candidates.

Q14.

Candidates should be given more information regarding the terminology that is used in questions. Where '**state**' or '**write down**' are used it is to indicate to candidates that the answer can reasonably be simply written down and that working does not need to be shown. There were a great number of scripts seen where integration was used in order to find the answers to parts (a) and (b). Although the correct answer to each of these parts was eventually arrived at, it was certainly not the most efficient way of doing so. The value of $P(X > 0)$ was usually found correctly to be 0.6 but again, far too many candidates

wasted time by using integration. Part (d) was only usually done correctly by the most able candidates. The vast majority of candidates usually perceived incorrectly that $P(X > 3.5)$ was required and so did not attempt to find $P(X < -3.5) + P(X > 3.5)$.

Q15.

Answers to part (a) were usually correct, although 4^2 was sometimes omitted or 4 was not squared. Some candidates could apparently not recall the formula for the circumference of a circle. Those candidates who could, invariably obtained 24π for its mean. The variance caused more trouble, often because π was not squared, or was omitted. In other solutions, the 2 was not squared. The correct use of brackets in expanding $(2\pi)^2$ seemed to have been forgotten by far too many candidates. Part (c)(i) was often answered correctly, but again there were some poor attempts at the expansion of $(X + 8)^2$. Part (c)(ii) caused considerable difficulty for many candidates because they were under the mistaken impression that $E(X^2)$ was the same as $(E(X))^2$. Hence they obtained 144π , rather than 145.2π , for which no marks were awarded.

Q16.

In part (a) it was very disappointing to see $P(T = 4) = 0.25$ in a large number of scripts. The correct answer of 'zero' was seen very infrequently.

In part (b) there were many candidates who correctly considered $\int_0^{\infty} \frac{t^2}{18} dt = 0.5$ and usually went on to gain full marks. However there were many other unsuccessful attempts to show that the median was 3. These mainly consisted of the substitution of $t = 3$ into one

of the expressions $\frac{t^2}{18}$ or $\frac{1}{4}(5-t)$, giving an answer of 0.5. This then led to a false deduction.

Many very good solutions were seen to part (c) despite the indifferent starts seen in parts (a) and (b). The most common approach was by integration methods with very few candidates using the geometry of the distribution to achieve the required result. Unfortunately, a few candidates thought that <3.5 meant that the limit to use was 3.45 or 3.49.

Many excellent attempts at the correct method to find the mean were seen in part (d). Unfortunately, some candidates could not cope with the fairly simple and straightforward integration.

Q17.

The majority of candidates showed a good understanding of this topic and so were able to score a good proportion of the marks available.

There were many very good sketches seen in part (a), although some candidates thought that $\frac{1}{54}t^2$ was the equation of a straight line. Parts (b)(i) and (ii) were well done by those candidates who used the areas of a rectangle and triangle to obtain their answers. Despite candidates being instructed to use their graphs in this part of the question, some candidates still used integration methods and so almost certainly wasted valuable time.

Part (b)(iii) was often attempted by use of $\int_0^{\infty} f(t) dt = 0.25$. Unfortunately, most candidates wrongly thought that this equated to the use of $\int_0^{\infty} \left(\frac{t^2}{54}\right) dt = 0.25$, and consequently gained no credit. The correct answer of $m = 3.5$ was rarely seen. There were many excellent solutions to part (c) with the correct answer of $\frac{4}{81}$ being seen on the majority of scripts.

Q18.

Most candidates made a good start with this question and the results for the mean and standard deviation of the exponential distribution were well known. The majority of candidates used the given formula to calculate the probability in part (b) and to find the median in part (c). A few attempted both these parts by integrating the function they had written down in part (a)(i) and this wasted time.

Q19.

In part (a) there were many excellent solutions, with the vast majority of candidates able to manipulate fractions to obtain the correct value of k . Unfortunately some candidates used inexact decimal calculations, which was unacceptable at this level when an exact value was required. Since the answer was given in this question a full and detailed method was essential for full marks to be awarded. Although there were many excellent solutions to

part (b), several candidates wrongly thought that $P(S < 3) = \int_0^3 f(s) ds$ where $\int_1^3 f(s) ds$ was required. Those candidates who attempted to find the distribution function invariably omitted the constant of integration. The correct form of the distribution function was:

$$F(s) = \begin{cases} 0 & s < 1 \\ \frac{1}{124}(s^3 - 1) & 1 \leq s \leq 5 \\ 1 & s > 5 \end{cases}$$

This then gave $F(3) = \frac{26}{124} = \frac{13}{62} = 0.210$.

Part (c) was not well attempted, with many candidates confusing the median with the mean and consequently evaluating $\int_1^5 sf(s) ds$ instead of using $\int_1^{\infty} f(s) ds = 0.5$, giving $m^3 = 63$ with a final answer of 4 minutes. However, some of those following the correct method lost an accuracy mark by not giving the answer correct to the nearest minute. As with part (b), some candidates thought that the lower limit was zero and consequently lost marks.

Q20.

Although this was not a particularly high scoring question, candidates as a body seemed to prefer it to questions set on previous papers which have required integration. A few

candidates were unaware that $\int_1^2 (x - 1.54)^2 f(x) dx$ is the variance. Instead they expanded

this integral and hence evaluated $E(X^2)$ for part (b)(iii). They then used this to evaluate the standard deviation in part (b)(ii).

Q21.

The proportion of candidates able to give a satisfactory answer to part (a) was much higher than that of those who could answer a similar question in January this year. Many candidates found the mean instead of the median in part (b). Most candidates picked up one or more marks in part (c), although some essentially started again instead of using the result in part (a). Only the best realised that Suggestion C was invalid as the probability density function would be negative for some values of x .

Q22.

Some candidates were thrown by this question which was in an unfamiliar form. Explaining why 0.75 was the mode caused particular problems. However, the stronger candidates scored well.

